

How much dissipation can a multiscale layer cascade carry?

Felipe Santibañez-Leal 

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github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Multiscale layer constructions produce finite-time blowup for several fluid equations: for the incompressible porous media equation, for the two-dimensional Boussinesq system, for three-dimensional Euler, and, with fractional dissipation $|\nabla|^\alpha$ of order $\alpha < (22 - 8\sqrt{7})/9$, for the forced Navier-Stokes equations. The exponent in the last of these is small, and it is natural to ask what limits it.

We give two upper bounds on the dissipation exponent such constructions can carry. The first is independent of every design constant: a cascade whose layers grow at rate $\sqrt{A}$ on a background gradient A cannot exceed $\alpha = 1/2$, and one whose layers grow at rate A cannot exceed $\alpha = 1$, both in the $|\nabla|^\alpha$ convention in which classical viscosity is $\alpha = 2$. Neither mechanism reaches classical viscosity, by factors of four and two respectively. The second bound applies to the smooth-forcing design used for the inviscid Boussinesq and Euler results: its correction hierarchy, seed rule and frequency-ratio rule cap the exponent at $\alpha \leq 1.08 \times 10^{-3}$, attained when exactly one derivative of the force is controlled, which is 86 times below the exponent already achieved with a force of finite regularity.

Both bounds follow from the amplitude system of the construction together with one additional ingredient, the effect of fractional dissipation on a single layer, which we derive and then verify against direct simulation of the nonlinear equations, including the effect of the spatial localization the constructions use. We also reconstruct the exponent budget behind the published hypodissipative threshold, identify the constraint that determines it, and show that the admissible frequency ratios form an interval that closes exactly at that threshold, so that cascades with geometrically growing frequencies are excluded at every positive dissipation.

1 The question

Let θ and u solve the forced two-dimensional Boussinesq system, or let u solve the forced three-dimensional Navier-Stokes equations with fractional dissipation. Multiscale constructions in the style introduced by Córdoba and Martínez-Zoroa build blowup by stacking oscillatory layers: each layer grows on the gradient deposited by its predecessors and hands a larger gradient to its successor, with infinitely many stages compressed into finite time.

Dissipation opposes this directly. A layer of frequency λ is damped at rate $\nu\lambda^{2\alpha}$ in the convention $-\nu(-\Delta)^\alpha$, while its growth rate is set by the background gradient alone. Frequency therefore buys gradient but not growth, and the question of how much dissipation a cascade tolerates becomes a question about the exponents relating amplitude, frequency and gradient. This paper answers it at two levels: for the mechanism classes (Section 3) and for one published design (Section 4).

Conventions. We write dissipation as $-\nu(-\Delta)^\alpha$, so classical viscosity is $\alpha = 1$; results quoted from the literature in the $|\nabla|^\alpha$ convention have twice our exponent, and we state which convention is in use whenever a number appears. The published hypodissipative threshold $(22 - 8\sqrt{7})/9 = 0.09267$ in the $|\nabla|^\alpha$ convention is 0.04633 in ours.

2 The layer mechanism and its dissipative extension

On a background that is affine near the origin, $u_{\text{old}} = D(t)x$ and $\theta_{\text{old}} = G(t) \cdot x$, a temperature wave and a vorticity wave sharing one phase $s = \lambda\zeta(t) \cdot x$, with $\vartheta = \Theta F(s)$, $\varpi = \Omega F'(s)$ and $\psi = \Omega P(s)/(\lambda^2|\zeta|^2)$, satisfy a closed system of amplitude equations,

$$\dot{\zeta} = -D^T \zeta, \quad \dot{\Theta} = -\frac{J\zeta \cdot G}{\lambda|\zeta|^2} \Omega, \quad \dot{\Omega} = \lambda \zeta_1 \Theta, \quad (1)$$

where $J(x_1, x_2) = (-x_2, x_1)$. The wave velocity is perpendicular to ζ while both wave gradients are parallel to it, so the wave does not advect itself and the ansatz is exact where the envelope is one and the profile is linear. In the frozen configuration $D = 0$, $G = -Ae_2$, $\zeta = re(\varphi)$, the growth rate is $\sqrt{A} \sin \varphi$: it carries no λ , because the frequency in $\dot{\Omega}$ cancels the reciprocal frequency in $\dot{\Theta}$. The gradient the layer deposits is governed by the amplification identity

$$\nabla \vartheta(0, t) = \lambda \Theta \zeta, \quad (2)$$

which is why a short wave can carry a small amplitude and a large gradient, and is what makes the cascade possible.

Dissipation. Adding $-\nu(-\Delta)^\alpha$ to both equations leaves the two geometric cancellations untouched and is diagonal on the phase, since $(-\Delta)^\alpha F(\lambda\zeta \cdot x) = (\lambda|\zeta|)^{2\alpha} F(\lambda\zeta \cdot x)$. The amplitude system therefore acquires one damping term per amplitude,

$$\dot{\Theta} = -\frac{J\zeta \cdot G}{\lambda|\zeta|^2} \Omega - \nu(\lambda|\zeta|)^{2\alpha} \Theta, \quad \dot{\Omega} = \lambda \zeta_1 \Theta - \nu(\lambda|\zeta|)^{2\alpha} \Omega, \quad (3)$$

with eigenvalues $-\nu(\lambda r)^{2\alpha} \pm \sqrt{A} \sin \varphi$ in the frozen configuration. Growth at frequency λ on background gradient A therefore requires

$$\nu(\lambda r)^{2\alpha} < \sqrt{A} \sin \varphi. \quad (4)$$

Equation (3) is the one ingredient of this paper that does not appear in the constructions themselves; Section 6 reports its verification against direct simulation of the nonlinear equations.

Two growth laws. The Boussinesq layers above grow at $\sqrt{A}$, a pendulum law: the amplitude pair obeys a second-order equation whose coefficient is the background gradient. The vortex layers of the hypodissipative Navier-Stokes construction instead grow at rate A_{n-1} , the background gradient itself, which is why its dissipation constraint reads $a = \alpha R$ in the exponent notation of Section 5 rather than carrying a square root. Both laws are treated below.

3 A bound independent of design constants

Theorem 3.1. *Consider a layer cascade with the following structure:*

- (A1) *each layer stops at amplitude $\lambda_q^{-1+\delta}$ with $\delta < 1$;*
- (A2) *it deposits a background gradient $A_{q+1} = \lambda_q^\delta$, by (2);*
- (A3) *frequencies and background gradients are related by $\lambda_q = A_q^p$ for a fixed p ;*
- (A4) *layers grow at rate A^γ with $\gamma = 1/2$ or $\gamma = 1$;*
- (A5) *a layer of frequency λ is damped at $\nu\lambda^{2\alpha}$, as in (3).*

Then $p\delta > 1$, and growth at every stage requires

$$\alpha \leq \frac{\gamma}{2p} < \frac{\gamma\delta}{2} < \frac{\gamma}{2},$$

that is $\alpha < 1/4$ for $\gamma = 1/2$ and $\alpha < 1/2$ for $\gamma = 1$. In the $|\nabla|^\alpha$ convention these ceilings are $1/2$ and 1 , against a classical viscosity of 2 .

Proof. By (A2) and (A3), $A_{q+1} = \lambda_q^\delta = A_q^{p\delta}$. The gradients must grow without bound for the cascade to produce blowup, so $p\delta > 1$; with $\delta < 1$ from (A1), which is required because the field itself stays bounded while its gradient diverges, this gives $p > 1/\delta > 1$.

By (A5) and (A4), stage q grows only while $\nu\lambda_q^{2\alpha} < cA_q^\gamma$ for a constant c independent of q . Taking logarithms and using (A3),

$$\log \nu + 2\alpha p \log A_q < \log c + \gamma \log A_q,$$

and since $A_q \rightarrow \infty$ the inequality can hold for all large q only if $2\alpha p \leq \gamma$, that is $\alpha \leq \gamma/(2p)$. At equality the condition reduces to $\nu < c$, which a small viscosity satisfies, so this step is not strict; strictness of the final bounds comes from $p > 1/\delta$, which gives $\alpha \leq \gamma/(2p) < \gamma\delta/2 < \gamma/2$. $\square$

In the variables of Section 4, where frequencies satisfy $\lambda_q = \lambda_{q-1}^Q$, the exponents are related by $Q = p\delta$, and the bound $\alpha \leq \gamma/(2p)$ reads $\alpha \leq \gamma\delta/(2Q)$, which for $\gamma = 1/2$ is the constraint $\alpha \leq \delta/(4Q)$ used there.

Corollary 3.2. *No cascade of either type reaches classical viscosity. The pendulum mechanism falls short by a factor of four and the vortex-stretching mechanism by a factor of two, uniformly in every constant of the design.*

The two ceilings differ by exactly the growth law: a layer that grows at the background gradient, rather than at its square root, gains a factor of two and nothing else in the argument changes. The published constructions sit well below their own class ceilings, the hypodissipative vortex-layer threshold by a factor 10.8 and the design of Section 4 by a factor 465.

Remark 3.3. Corollary 3.2 is consistent with a choice visible in the literature: the construction aimed at the viscous case does not use a layer cascade, but a self-similar core in which the scale-local Reynolds number stays of order one.

4 A bound for the smooth-forcing design

The inviscid Boussinesq and Euler constructions with smooth forcing fix their design as follows. The frequencies satisfy $\lambda_q = \lambda_{q-1}^{Q_q}$ with $Q_q = Q^* + q$ and $Q^* \geq 200$; the number of derivatives of the force controlled uniformly is k_q , constrained by $120k_q \leq Q_q$; the localization error is cancelled through $J = 2k + 8$ correction levels, each gaining a factor $\Pi^5 \lambda^{-(1-\delta)}$ where $\Pi \leq \lambda^{5/Q}$; and the seed is chosen so that the remaining force stays below λ^{-6} . With the published amplitude rule a layer stops at $\lambda^{-7/8}$, so the margin of (A1) is $\delta = 1/8$.

Theorem 4.1. *Under that design, carrying dissipation $-\nu(-\Delta)^\alpha$ requires*

$$J\left[(1-\delta) - \frac{5}{Q}\right] \geq k+6 \quad \text{and} \quad \alpha \leq \frac{\delta}{4Q},$$

the first from the force budget and the second from (4). With $Q = 120k$, the smallest ratio the design permits, these give

$$\delta \leq 1 - \frac{k+6}{2k+8} - \frac{1}{24k}, \quad \alpha \leq \frac{1}{480k} \left(1 - \frac{k+6}{2k+8} - \frac{1}{24k}\right),$$

whose maximum over $k \geq 1$ is attained at $k = 1$:

$\alpha \leq 5.38 \times 10^{-4} \quad (1.08 \times 10^{-3} \text{ in the } |\nabla|^\alpha \text{ convention}), \quad \delta \leq 0.2583, \quad Q = 120.$

Proof. Each correction level gains the factor $\Pi^5 \lambda^{-(1-\delta)}$ and there are J of them, so the residual force carries the exponent $-J[(1-\delta) - 5/Q]$ in λ after using $\Pi \leq \lambda^{5/Q}$. Controlling k derivatives of a field oscillating at frequency λ costs λ^k , and the design requires the result to stay below λ^{-6} , which is the first display. The second is (4) with $A_q = \lambda_{q-1}^\delta$ and $\lambda_q = \lambda_{q-1}^Q$, taking the insertion angle of order one, which is the most favourable case. Eliminating δ and setting $Q = 120k$ gives the displayed formula; it is decreasing in k , since the bracket increases to $1/2$ while the prefactor decreases like $1/k$, so the maximum is at $k = 1$. $\square$

k	$J = 2k + 8$	$Q = 120k$	$\delta \leq$	$\alpha \leq$
1	10	120	0.2583	5.38×10^{-4}
2	12	240	0.3125	3.26×10^{-4}
3	14	360	0.3433	2.38×10^{-4}
10	28	1200	0.4244	8.84×10^{-5}
1000	2008	120000	0.4990	1.04×10^{-6}

The bound is 86 times below the exponent 0.04633 already reached with a force in $L_t^1 C_x^{1,\epsilon} \cap L_t^\infty L_x^2$, and the shortfall does not close by retuning: at that exponent the ratio rule cannot buy even one derivative. Smoothness of the force and fractional dissipation are purchased from a single budget in this design, with the frequency ratio as the currency: 120 units of ratio per derivative, and the exponent inversely proportional to the ratio.

Remark 4.2. Three quantities enter the proof, so a dissipative version of this design must move one of them: decouple the correction levels from the frequency ratio, so that Q need not grow with k ; raise the amplitude margin beyond $1/2$, which the hierarchy as stated forbids; or change the growth law, since the factor four in $\alpha \leq \delta/(4Q)$ comes from the rate $\sqrt{A}$ and becomes a factor two for vortex stretching.

5 The exponent budget behind the published threshold

The hypodissipative Navier-Stokes construction writes every quantity as a power of one base: the frequency of layer n is $M_n = N^{R^n}$, the amplitude $A_n = N^{aR^n}$, the localization scale $L_n = N^{bR^n}$, and $s > 0$ is the Hölder margin of the force. Its force estimates give four constraints:

	origin	constraint
D	dissipation	$a = \alpha R$
S	self-interaction	$2a - a/R + b + s - 1 \leq 0$
L	localization	$b \geq a + 1/R + s$
O	outer velocity on the inner layer	$a + 2/R + s + 1 - 3b \leq 0$

Proposition 5.1. *With **D** and **S** saturated, **O** reduces to $4s < 2 + 3\alpha - 7\alpha R - 2/R$, whose maximum over the frequency ratio is at $\alpha R^2 = 2/7$. At that optimum $s(\alpha) = (2 + 3\alpha - 2\sqrt{14\alpha})/4$, and $s > 0$ is equivalent to $9\alpha^2 - 44\alpha + 4 > 0$, whose smaller root is $(22 - 8\sqrt{7})/9$.*

The optimizing ratio $R = \sqrt{2/(7\alpha)}$ is the one the construction uses, and no coefficient in the chain is fitted. At the threshold the localization constraint is inactive, with margin $\frac{1}{2}(\sqrt{2\alpha/7} - \alpha) = 0.0350$; a budget that saturates **L** instead and omits **O** yields $\alpha R^2 = 1/3$ and the weaker bound $5 - 2\sqrt{6} = 0.10102$, the value given by an order-of-magnitude estimate in the source. The difference between the two is precisely which constraint is active.

Proposition 5.2. *A positive force margin requires $7\alpha R^2 - (2 + 3\alpha)R + 2 < 0$, so the admissible ratios form the interval with endpoints $R_{\pm} = [(2 + 3\alpha) \pm \sqrt{9\alpha^2 - 44\alpha + 4}]/(14\alpha)$. The discriminant is the polynomial of Proposition 5.1, so the interval closes to a single point exactly at the threshold, and that point is $\sqrt{2/(7\alpha)}$. Moreover $R_- > 1$ for every $\alpha > 0$, and at $R = 1$ the margin equals $-\alpha$.*

α ($ \nabla ^\alpha$)	admissible R	optimizing R
0.001	(1.002, 285.1)	16.90
0.01	(1.021, 27.98)	5.345
0.05	(1.143, 5.000)	2.390
0.09267	{1.7559}	1.7559
0.10	empty	none

Corollary 5.3. *Cascades with geometrically growing frequencies, the case $R = 1$, are inadmissible at every positive dissipation: localization requires $b \geq a + 1 + s$ while self-interaction permits only $b \leq 1 - a - s$, forcing $\alpha + s \leq 0$. Super-geometric frequency growth is therefore necessary, and more of it is necessary as the dissipation rises, until at the threshold a single ratio remains.*

Propositions 5.1 and 5.2 are verified in exact arithmetic by computer algebra in the accompanying repository.

6 Verification of the dissipative extension

Theorems 3.1 and 4.1 rest on (3), which is derived rather than taken from the constructions. It was tested against direct pseudospectral simulation of the nonlinear two-dimensional Boussinesq system in double precision with two-thirds dealiasing.

The amplitude system. On the stratification $\theta = -(A/\lambda_0)\sin(\lambda_0 x_2)$ the measured peak growth rate matches $\sqrt{A}\sin\varphi$ to 4×10^{-5} relative at two insertion angles; the predicted independence of the rate from λ holds to 2.7×10^{-4} across an eightfold range; and the damping term of (3) tracks the measured rate to 2.6×10^{-4} absolute across $\alpha \in \{1/4, 1/2, 1\}$ and $\nu \in \{10^{-5}, 10^{-4}\}$.

The cascade premise. On a two-scale background that is an exact steady state, with drift 1.45×10^{-15} , the local growth rate of a fine wave follows the total gradient at correlation 0.99945 and the base stratification alone at 0.212: a layer responds to the accumulated low-frequency gradient rather than to the base.

The growth, steering and holding cycle. The control that returns a grown layer to rest rotates the background gradient and the phase direction together, so that the laboratory component ζ_1 changes sign while $J\zeta \cdot G$ and $|\zeta|$ are preserved. In the co-rotating frame it becomes a rotating gravity direction, with the stratification held by a low-frequency force. On a background whose gradient is affine to fourth order the cycle reproduces the amplitude system to 3.06×10^{-6} in the steering gain and returns the vorticity amplitude to 3.67×10^{-4} of its peak, where it holds to 0.36% over three growth times; on a plain sinusoidal background the same quantities are 1.80×10^{-2} and 1.57×10^{-2} , the difference being the curvature of the background over the layer's support. Under dissipation the cycle factorizes as e^{-dt} times the inviscid cycle with $d = \nu\lambda^{2\alpha}$: a holding interval decays at d to within 0.97%, and the rescaled viscous run reproduces the inviscid one to 0.24%.

Localization. Since the constructions localize each layer, (3) is applied to a band rather than to a single frequency. For an envelope of radius ℓ on a wave of frequency λ the relative error of the damping law is

$$\frac{\|(-\Delta)^\alpha(g e^{i\lambda\zeta \cdot x}) - (\lambda|\zeta|)^{2\alpha}g e^{i\lambda\zeta \cdot x}\|}{\|(\lambda|\zeta|)^{2\alpha}g\|} \approx \frac{3.6\alpha}{\ell\lambda},$$

first order in the bandwidth ratio, with measured slope -0.9863 , and depending on that ratio alone: configurations sharing $\ell\lambda$ but splitting it differently between frequency and envelope agree to four digits. The coefficient is linear in α , as the symbol expansion predicts. At the separations the constructions use, $\ell_q^{-1} = \lambda_{q-1}^3$ against $\lambda_q = \lambda_{q-1}^{Q_q}$ with $Q_q \geq 201$, the error is $\lambda_{q-1}^{3-Q_q}$, so localization does not affect the bounds above.

A property of the mechanism in an unforced periodic setting. The stratification these layers grow on is Rayleigh-Taylor unstable at rate $\sqrt{A}$, while a layer inserted at angle s grows at $\sqrt{A}\sin s$. Perturbations at more favourable angles therefore gain $1/\sin s$ times as many e-foldings as the layer. The constructions are unaffected, since each layer is localized and the force controls the exterior, but any realization without that control must keep its schedule short enough for the layer to stay ahead.

7 The status of the announced results

The two September 2026 announcements are accompanied by machine-checked certificates. For the Navier-Stokes and Euler development, the headline statements match the forced alternatives of the official problem statement clause by clause; the development builds with no incomplete proofs and its four headline theorems depend only on the three standard axioms; and every

constant in the closure of both root modules replays through the Lean kernel from an empty environment at the pinned toolchain version. The reference statement is adapted from an independently maintained collection of formalized conjectures, so the definitions that determine whether the formalized statement is the intended one predate the claim. None of this speaks to the mathematics outside the formalization, and kernel replay detects environment manipulation rather than serving as an external verifier.

8 Scope

The bounds of Sections 3 and 4 are necessary conditions for growth in the layer mechanism, derived from the amplitude system, the amplification identity and the dissipative term (3). They are exponent statements: constants, logarithmic factors and the corrections that make the localized ansatz exact are not tracked, and the published inviscid results are not affected by them, since those results carry no dissipation. The mechanism is due to Córdoba and Martínez-Zoroa, extended to smooth forcing by Alpogé and Buckmaster; nothing in this paper establishes blowup for any equation or improves any published threshold.

An earlier version of this paper stated that the value $p = 11/4 + \sqrt{7}$, obtained by evaluating the relation $\alpha_c = 1/(4p)$ at the published threshold, carried information about that threshold. It does not: the relation is constraint **D** saturated, which the construction saturates at every α , so the identity holds along the whole family, and its algebraic form is automatic because $\alpha_0 \in \mathbb{Q}(\sqrt{7})$ and the norm of $22 - 8\sqrt{7}$ is a perfect square. That statement is withdrawn.

Data and code

The numerical data, the simulations, the exact-arithmetic verifications of Propositions 5.1 and 5.2, and the computations behind Theorems 3.1 and 4.1 are public at [the project repository](#).

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